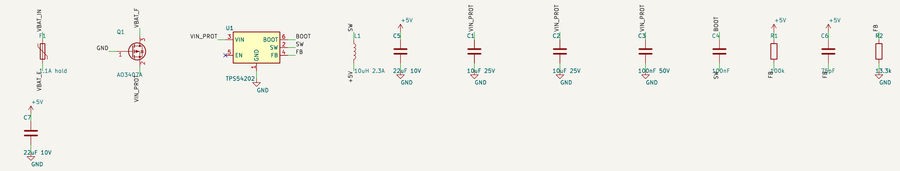
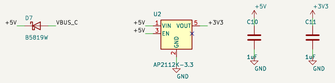
Bay supply: reverse protection, polyfuse, 5 V buck (TPS54202)



JR module bay connector



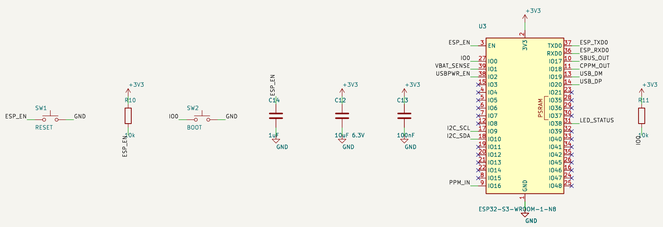
USB-C 5 V for bench testing (Schottky OR into the 5 V rail)



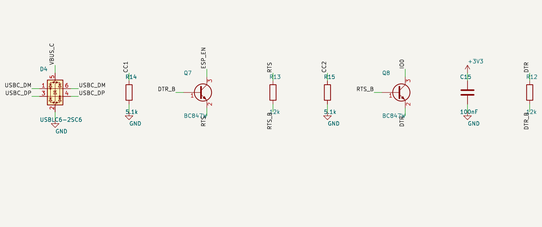
USB-A host port: VBUS switch, polyfuse, bulk capacitance, ESD



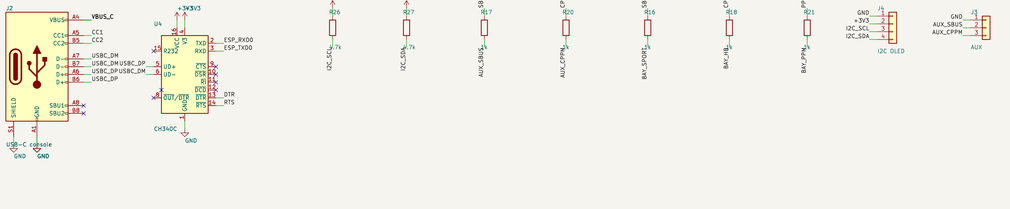
ESP32-S3-WROOM-1 (USB host, SBUS/CPPM outputs)



USB-C console and flashing: CH340C, auto-reset



Outputs: bay S.PORT/heartbeat, AUX header, trainer jack, I2C display



Indicators and battery sensing

Power flags (ERC)

esp32-joystick2trainer JR bay module, rev A. Generated from hardware/scripts/design.py.
JR bay pins: 1 PPM, 2 heartbeat, 3 VBAT, 4 GND, 5 S.PORT (verify on the radio; JP1/JP2 select GND/heartbeat on pins 2/4).
Bay supply 2S (6.0–8.4 V) only: Q1 Vgs rating. USB-C 5 V powers the board when the bay supply is absent (no back-feed).
Firmware pins: SBUS GPIO17, CPPM GPIO18, USB host GPIO19/20, LED GPIO38, VBUS enable GPIO21, I2C GPIO8/9, VBAT sense GPIO1.

esp32-joystick2trainer (MIT firmware, CERN-OHL-P-2.0 hardware)
Sheet: /
File: j2t-module.kicad_sch
Title: esp32-joystick2trainer JR bay module
Size: A1 | Date: 2026-11-06 | Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 1/1